

## CONDITIONAL EINSTEIN BRANCH TO NEWTON

### EINSTEIN'S EQUATION

$$G_{\alpha\beta} + \Lambda g_{\alpha\beta} = 8\pi G \langle T_{\alpha\beta} \rangle$$

weak field, slow motion

### POISSON'S EQUATION

$$\nabla^2 \Phi = 4\pi G \rho$$

point mass

### GRAVITATIONAL POTENTIAL

$$\Phi = -GM/r$$

$$F = -m\nabla\Phi$$

Geodesic eq.

### NEWTON'S LAW

$$F = -GMm/r^2$$

*Given the Einstein branch, each step is a standard limit*